

Deepanshu Singh

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EDUCATION

Indian Institute of Technology Madras

Chennai, India

B.S. in Data Science and Applications

Aug. 2023 – Jul. 2027 (Expected)

- **Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Probability & Statistics, Linear Algebra, Data Structures & Algorithms, Statistical Modelling

PROJECTS

BLaIR: Dense Product Retrieval

Jan. 2026 – Present

PyTorch, BERT, BGE, E5, FAISS IndexFlatIP, InfoNCE, McNemar, Bonferroni, GitHub Actions CI/CD

- Benchmarked **11 systems** (BM25, zero-shot BERT, SBERT, E5, BGE, bi-encoder, dual-encoder, hard-neg, hybrid RRF, BGE fine-tuned, cross-encoder reranker) on 56,921 Amazon Electronics products; implemented FAISS IndexFlatIP vector search and InfoNCE contrastive loss — directly replicating the **Amazon Search** problem of bridging vocabulary mismatch between customer reviews and product descriptions
- Selected BGE-base backbone (highest zero-shot NDCG@10 = 0.0793); domain fine-tuning on 80k pairs achieved **NDCG@10 = 0.0995 — 378% over BM25** (McNemar $p < 0.01$, Bonferroni-corrected, 9,972 queries); outperformed **SBERT pre-trained on 1B+ pairs** ($p = 0.0016$) — domain adaptation beats general-purpose pretraining; full end-to-end research to prototype to production pipeline
- Converted single-point observation into empirical **data-scale finding**: trained both architectures at 20k/50k/100k pairs; shared-weight BiEncoder beats DualEncoder at $\leq 100k$ pairs ($p = 0.0086$) with gap closing as data grows; 6-point scaling curve confirms BLAIR's data-scale hypothesis on large-scale datasets
- **43 unit/integration tests + GitHub Actions CI/CD**; 11 McNemar tests (Bonferroni-corrected); 3-seed stability ($\text{std} = 0.0004$ across MPS+CUDA); temperature sweep ($\tau = 0.01\text{--}0.2$); pooling ablation; honest failure diagnosis — RRF and cross-encoder reranker both diagnosed as domain-mismatch failures; t-SNE shows 70.2% pair-distance reduction after fine-tuning

H&M Personalized Fashion Recommendation System

Sep. 2025 – Jan. 2026

PyTorch, LightGBM, FAISS, Polars, InfoNCE, SHAP, Bootstrap CI, Python

- Architected an industrial **5-stage pipeline** (time-decayed Popularity $\rightarrow$ ALS $\rightarrow$ InfoNCE Two-Tower $\rightarrow$ RRF Fusion $\rightarrow$ LightGBM LambdaMART) on **90,246 users, 84,220 items, 1.8M real H&M interactions**; achieved **NDCG@12 = 0.0105** and **Recall@12 = 0.0120** (**+140.0% lift over baseline**, $p < 0.0001$, 1,000-sample bootstrap CI) with zero temporal leakage
- Engineered a **40-feature ranking pipeline** (7d/30d velocity, price affinity, retrieval ranks) reducing scoring from $O(N \cdot M)$ to $O(K \log N)$ via FAISS MIPS with **<15 ms P95 latency**; expanded catalog coverage to **22.92% (1,073 $\times$ over baseline)**; **54 tests + GitHub Actions CI/CD**

ResNet-34 from Scratch & Mechanistic Interpretability

Jun. 2025 – Aug. 2025

PyTorch, Computer Vision, Optimization Dynamics, Grad-CAM, Amazon Berkeley Objects, CI/CD

- Engineered a 21.3M-parameter ResNet-34 from scratch using raw PyTorch primitives (autograd-safe residual projections, zero-bias Conv-BN, Kaiming Normal init); achieved **93.37% $\pm$ 0.27% on CIFAR-10** across 6 multi-seed GPU runs and scaled to **44,866 Amazon Berkeley Objects images (50 classes)**, reaching **81.37% top-1 accuracy**; verified by **172 unit/integration tests + GitHub Actions CI/CD**
- Diagnosed optimization degradation by profiling layer-wise gradient norms, discovering **gradient magnitude inversion in Plain-34 (0.25 $\times$ ratio vs. 1.5 $\times$ in ResNet-34)** driving a **+18.4pp accuracy gap**; built hook-based Grad-CAM validated via Adebayo et al. parameter randomization ($|\rho| = 0.031$) to isolate **background bias** as the primary e-commerce catalog classification failure mode

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, SQL, Bash

Machine Learning & Deep Learning: Machine Learning, Deep Learning, Neural Networks, Supervised/Unsupervised Learning, Reinforcement Learning, Representation Learning, Contrastive Learning (InfoNCE), Transfer Learning, Model Fine-tuning, Large-Scale ML Systems, End-to-End ML Pipelines, Research to Production

NLP & Information Retrieval: NLP, Natural Language Understanding (NLU), Dense Retrieval, Semantic Search, Bi-Encoder, Dual-Encoder, BM25, FAISS, Transformers, BERT, BGE, E5, SBERT, Word Embeddings, Sentence Embeddings, Vector Search, Hugging Face, Text Classification

Computer Vision: CNNs, ResNet, Grad-CAM, Image Classification, Object Recognition, Gradient Visualization, Model Interpretability, Data Augmentation

Recommendation Systems: Collaborative Filtering, ALS Matrix Factorization, Two-Tower Models, LambdaMART, Learning-to-Rank, Candidate Generation, Re-ranking, Personalization, SHAP Attribution, Implicit Feedback

Frameworks & Libraries: PyTorch, LightGBM, Scikit-Learn, NumPy, Pandas, Polars, Matplotlib, Seaborn, HuggingFace Transformers, FAISS

Experimentation & Evaluation: A/B Testing, McNemar's Test, Bootstrap CI, Bonferroni Correction, Statistical Significance Testing, Ablation Studies, Error Analysis, Offline Evaluation, NDCG, MRR, Recall

MLOps & Tools: Git, GitHub Actions CI/CD, Docker, Linux, Unit Testing, Integration Testing, Reproducible Research, Data Pipelines, Feature Engineering, Experiment Tracking

ACHIEVEMENTS & CERTIFICATIONS

Amazon ML Summer School 2026 | Selected from IIT Madras; conducted by Amazon Applied Scientists — Machine Learning, Deep Learning, NLP, Large-Scale ML Systems

Executive PG Certification in Data Science & AI | iHUB DivyaSampark, IIT Roorkee in association with Intellipaath

Competitive Programming | 250+ problems across LeetCode, GeeksforGeeks & HackerRank — arrays, graphs, trees, dynamic programming, algorithmic complexity